

ABKD: PURSUING A PROPER ALLOCATION OF THE PROBABILITY MASS IN KNOWLEDGE DISTILLATION VIA α - β -DIVERGENCE — A REPRODUCIBILITY STUDY

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ABSTRACT

Reproducibility is essential to scientific research. In this work, we attempt to reproduce several claims from the ICLR 2025 paper “ABKD: Pursuing a Proper Allocation of the Probability Mass in Knowledge Distillation via α - β -Divergence” using the authors’ published code. Our results are partially consistent with those in the paper, but differences remain. We also introduce an additional experiment to extend the analysis.

1 INTRODUCTION

Knowledge Distillation (KD) typically relies on Forward or Reverse KL Divergence (FKLD/RKLD), which often struggle to balance the focus on hard samples and student confidence. To address this, Wang et al. (2025) proposed ABKD, a generalized framework leveraging α - β -Divergence to smoothly interpolate between these extremes.

Our team attempted to reproduce the core claims regarding the vision tasks in Section 5.2 of Wang et al. (2025). Specifically, we aim to verify that:

- Without modifying the distillation objective, methods that more effectively utilize the teacher distribution knowledge (e.g., DKD, TTM, and LSD) generally outperform vanilla KD. However, their scores fall short in certain scenarios.
- Orthogonal to existing methods, the ABKD framework can select more suitable distillation objectives for specific teacher-student pairs, showing competitive or superior results under the standard training-evaluation paradigm.

Our Contributions: In this work, we first successfully reproduce the standard vision tasks to validate the baseline claims. Furthermore, as an **extension**, we challenge the robustness of the ABKD framework by shifting from a balanced dataset to a Long-Tailed distribution (CIFAR-100-LT). We analyze how α and β reallocate probability mass from head to tail classes, showing that ABKD reduces teacher bias and improves few-shot generalization under severe data imbalance.

2 EXPERIMENT WITH SUPPLIED CODE

2.1 EXPERIMENT SETUP

All experiments were conducted using the authors’ code from GitHub on the Google Colab platform, utilizing an **NVIDIA T4 GPU**. We strictly follow the settings in Wang et al. (2025) for all training settings, datasets, and model configurations to ensure a fair and reproducible comparison.¹

Methods. Our experimental design is divided into two phases. First, we establish robust baselines by reproducing four representative knowledge distillation methods: Vanilla KD, DKD, LSD, and TTM. Second, we validate the framework’s orthogonality and plug-and-play nature by replacing the standard distribution matching loss with the proposed α - β divergence, obtaining ABKD, ABDKD, ABLSD, and ABTTM.

¹Each run took ~ 4 hours on a Colab T4 GPU (1.19 units/hr). The observed ~ 3 GB GPU utilization indicates a severe CPU data-loading.

2.2 REPRODUCTION OF BASELINES

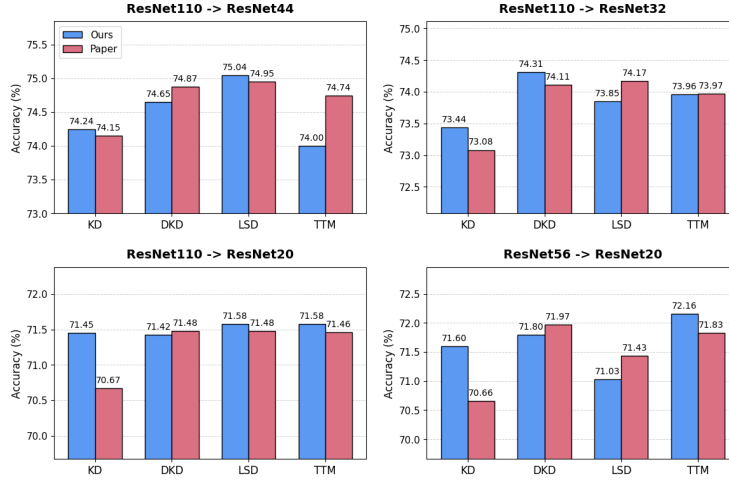
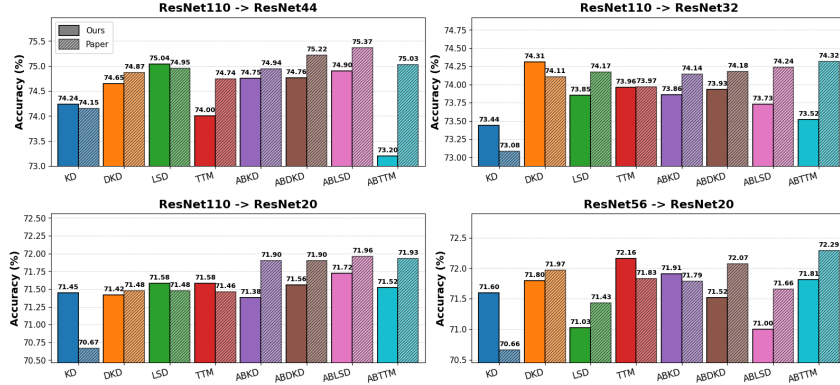


Figure 1: Comparison of top-1 accuracy of baselines

As illustrated in Figure 1, advanced distillation methods utilizing structured knowledge (e.g., DKD, LSD, TTM) generally outperform Vanilla KD, but their performance is highly unstable.

However, the gains are not consistent. For instance, in the ResNet110 $\rightarrow$ ResNet44 setting, TTM (74.00%) underperforms KD (74.24%). Furthermore, under extreme capacity gaps (e.g., ResNet110 $\rightarrow$ ResNet20), the improvements largely vanish, with DKD and Vanilla KD showing comparable performance.

2.3 REPRODUCTION OF α - β FRAMEWORK

Figure 2: Comparison of top-1 accuracy of baselines and α - β framework

The original paper claims that the AB-Divergence framework is orthogonal and seamlessly enhances existing KD methods. Based on Figure 3, our reproduction partially challenges this claim.

Improvement over Vanilla KD. The benefits of AB-Divergence on standard KD are consistent and significant. For example, KD improves from 74.24% to 74.75% (+0.51%) in ResNet110 $\rightarrow$ 44, from 73.44% to 73.86% (+0.42%) in ResNet110 $\rightarrow$ 32, and from 71.60% to 71.91% (+0.31%) in ResNet56 $\rightarrow$ 20. This suggests that AB-Divergence effectively rebalances probability mass and alleviates bias in Vanilla KD, leading to improved utilization of dark knowledge.

Conflicts with Advanced Methods. In contrast, the integration of AB-Divergence with more advanced KD methods often leads to performance degradation. For instance, ABDKD decreases performance by 0.38% (74.31% $\rightarrow$ 73.93%) in ResNet110 $\rightarrow$ 32 and by 0.28% in ResNet56 $\rightarrow$ 20. Similarly, ABTTM consistently underperforms standard TTM across all settings, with a notable drop of 0.8% (74.00% $\rightarrow$ 73.20%) in ResNet110 $\rightarrow$ 44.

These results indicate that the orthogonality claim does not always hold. We argue that superimposing the explicit AB-Divergence may introduce conflicts with the original optimization objectives, potentially leading to over-regularization ².

2.4 ISSUES WITH THE SUPPLIED CODE AND DOCUMENTATION

During our reproduction process, we identified several critical issues regarding the official repository’s code quality, documentation, and environmental compatibility:

- **Unverifiable Baseline Metrics:** The authors appear to have directly copied certain baseline results from other literature. However, upon investigating the cited references, the empirical data for specific configurations (e.g., ResNet110 $\rightarrow$ ResNet44) is not explicitly reported. This lack of primary data sources makes verifying the absolute accuracy of the original baseline claims impossible.
- **PyTorch 2.6 Compatibility Issue:** Loading the provided pre-trained teacher models fails in modern environments (e.g., Google Colab with PyTorch ≥ 2.6). For security reasons, PyTorch 2.6 sets `weights_only=True` by default in `torch.load()`, which intercepts and blocks the serialized objects embedded in the authors’ model files. A manual override (`weights_only=False`) is required to bypass this.
- **Flawed Example Scripts:** The provided bash scripts contain logical errors. For instance, executing the official example `bash train_ls.sh 1.0 1.0 0 0 ResNet56 ResNet20 0` initializes $\alpha = 0.0$. Since the underlying code only implements a safety fallback for $\beta = 0$, this specific configuration triggers an unhandled `ZeroDivisionError`.
- **Implementation Bug in TTM.py:** There is a fatal runtime bug in the `TTM.py` script. At line 11, the code attempts to access an undefined attribute `ab.gamma`, which immediately causes the program to crash. This is likely an uncorrected typo from the authors’ development phase.
- **Incomplete Hyperparameter Specifications:** The repository lacks explicit and comprehensive parameter configurations for all baseline scripts. Providing only partial argument examples introduces unnecessary guesswork, hindering precise and deterministic reproduction of the reported tables.

3 EXPERIMENT WITH OUR IMPLEMENTATION

Our code is committed at <https://github.com/BeckY824/DP-DL-Reproduction-A-BKD/tree/master>.

3.1 EXPERIMENT MOTIVATION AND SETUP

In our initial attempts to reproduce the experiments on the full CIFAR-100 dataset, we found the computational cost and time requirements to be prohibitively high for our available resources. To address this practical constraint while still rigorously exploring the efficacy of distillation methods, we adjusted our dataset configuration by switching to the CIFAR-100-LT dataset. This approach is inspired by the experimental settings in He et al. (2021).

The environment remains consistent with our previous setups, focusing exclusively on the ResNet-110 (Teacher) to ResNet-44 (Student) architecture pairing. All training is conducted on Google Colab using an NVIDIA T4 GPU³.

To construct the CIFAR-100-LT dataset, we set the imbalance factor to 0.01, yielding a standard Imbalance Ratio (IR) of 100 ($N_{\max} = 500, N_{\min} = 5$). For evaluation, we strictly use the original balanced CIFAR-100 test set, ensuring a fair and consistent performance measurement that aligns with our previous baselines.

For comprehensive comparison, we additionally incorporate the **DiVE** (Distilling Virtual Examples) method proposed in the reference paper, along with a standard **CE** (Cross-Entropy) baseline. Finally, to ensure the reliability of our reproduction results, each experiment in this section is repeated across **five** independent trials.

²Due to computational constraints, each experiment in our study was conducted only once. In contrast, the original work may report results averaged over multiple runs, providing more stable estimates. Therefore, our findings should be interpreted with caution, and further validation with repeated trials is required.

³By using the LT dataset, each run was reduced to approximately 1 hour on a Colab T4 GPU (1.19 compute units/hr) with an observed GPU memory utilization of ~ 3 GB.

3.2 EVALUATION OF BASELINES

Since TTM performed poorly in the previous experiment, we did not repeat it multiple times in order to save computational resources. The obtained results are consistent with our expectations, showing that it is not competitive with the other methods.

We can observe from Table 1 that the distillation-based methods achieve significantly better performance than both the baseline CE and the DiVE method from the reference paper, in terms of both top-1 accuracy and average accuracy.

Methods	Imbalance Ratio = 0.01	
	Top-1 Acc	Avg Acc
CE	38.35	-
DiVE	45.35	-
KD	53.29	52.63
DKD	56.45	55.77
TTM	47.79	-
LSD	57.43	57.29

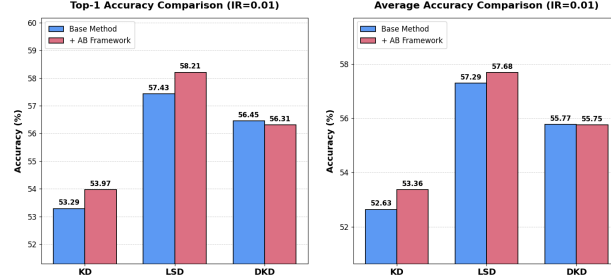


Table 1: Baseline performance on CIFAR-100-LT

Figure 3: Comparison of top-1 accuracy of baselines and $\alpha - \beta$ framework

3.3 EVALUATION OF THE $\alpha - \beta$ FRAMEWORK

Similarly, we evaluate all distillation methods under the AB framework (except for TTM) in Figure 3. We do not adopt the exhaustive parameter search strategy used in the original paper, as it is computationally expensive. Instead, we first select three representative settings for α and β : $[0.9, 0.1]$, $[0.5, 0.5]$, and $[0.1, 0.9]$. We then choose the most evidently optimal pair of α and β for repeated experiments. Empirically, we find that the optimal setting for all distillation methods is $[0.5, 0.5]$.

Secondly, we observe that the AB framework can improve the performance of the original distillation methods, particularly for KD and LSD. However, this improvement does not always hold, as it is not observed in the case of DKD.

4 CONCLUSION

From the selected experiments, we find that the AB framework can indeed improve the baseline performance of distillation methods to a certain extent, and this holds in most cases.

From an innovation perspective, the contribution of this paper is relatively modest. It does not introduce a fundamentally new algorithm; instead, it focuses more on empirical engineering practices and extensive hyperparameter tuning. Consequently, the overall accuracy improvements are relatively marginal.

However, it still provides a valuable perspective as an alternative direction for improvement. To some extent, it reduces the need to exhaustively evaluate all possible configurations, but at the same time, it may introduce additional computational overhead.

ACKNOWLEDGMENTS

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